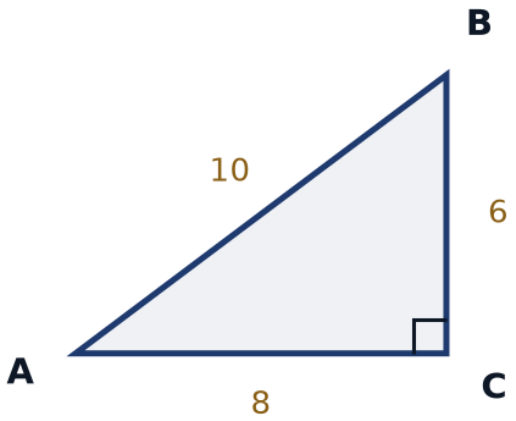
Calculus

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.



2 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

3 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

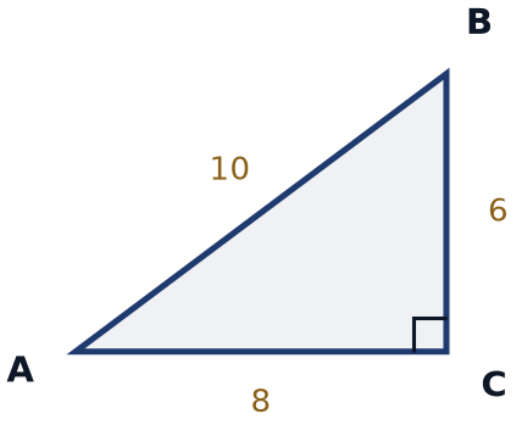
4 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

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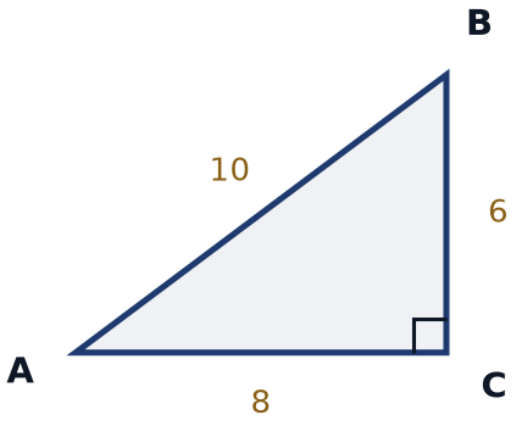
5 [Foundational] • Target: The unit circle and angles of any magnitude

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



6 [Foundational] • Target: The unit circle and angles of any magnitude

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



7 [Foundational] • Target: The unit circle and angles of any magnitude

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

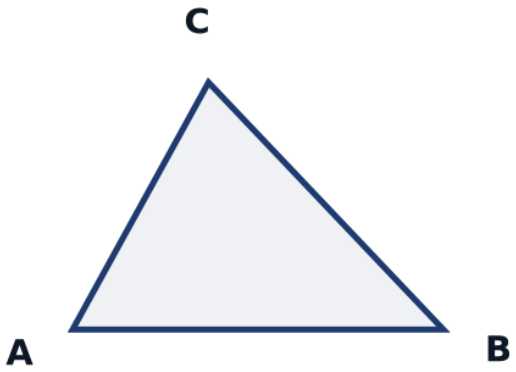
8 [Intermediate] • Target: The unit circle and angles of any magnitude

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

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9 [Intermediate] • Target: Trigonometric identities

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(2)$.



10 [Intermediate] • Target: Trigonometric identities

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

11 [Intermediate] • Target: Trigonometric identities

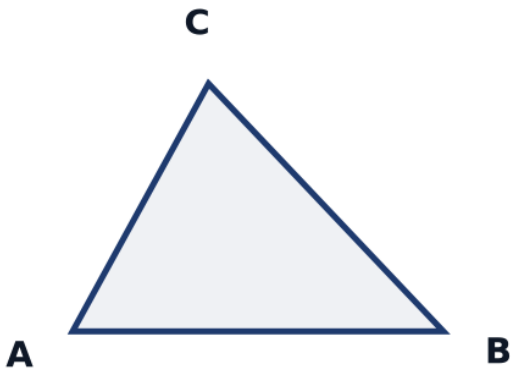
A data set recorded for trigonometric identities consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

12 [Intermediate] • Target: Trigonometric identities

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(3)$.

13 [Intermediate] • Target: Trigonometric identities

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



14 [Intermediate] • Target: Trigonometric identities

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(4)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

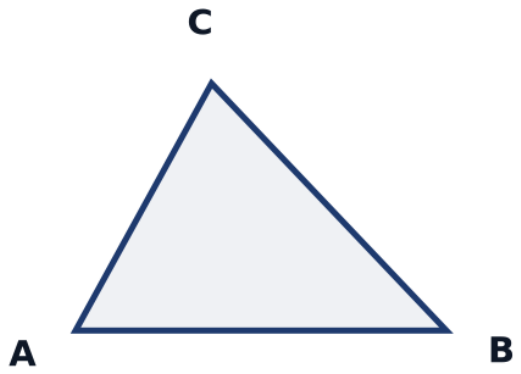
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15 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 4.01 Statistics & Probability

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

16 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(5)$.

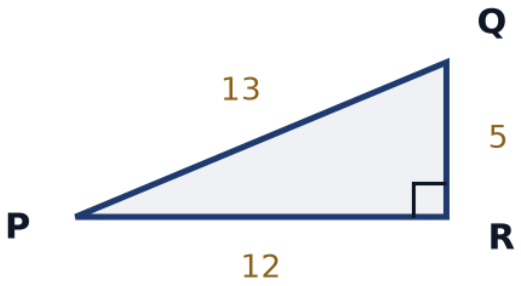


17 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 1.01 Number & Algebra

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

18 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 2.01 Functions

For the model $f(x) = 6x + 9$ in solving trigonometric equations 1, evaluate $f(2)$.



19 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 3.01 Geometry & Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

20 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for solving trigonometric equations 1 consists of the four values $\{17, 21, 25, 31\}$.

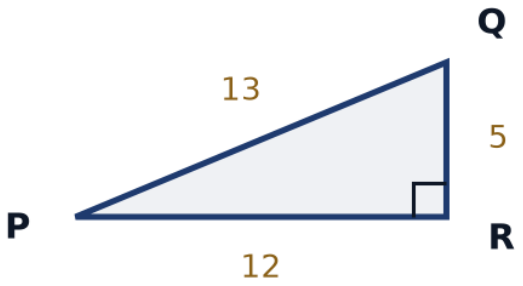
What is the arithmetic mean? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exact values in right-angled triangles

In ΔABC studying exact values in right-angled triangles, $m\angle A = 35^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 100°

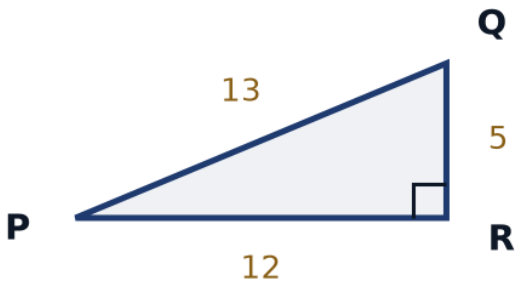
b. 110°

c. 92°

d. 145°

22 [Foundational] • Target: Exact values in right-angled triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?



a. 9 cm

b. 12 cm

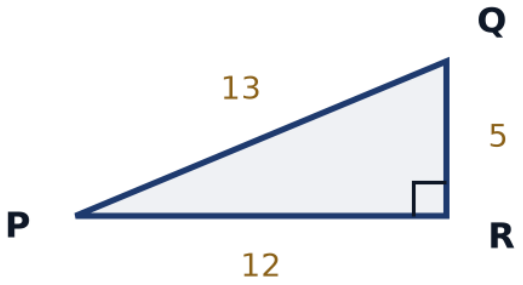
c. 16 cm

d. 36 cm

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23 [Foundational] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?



a. 13

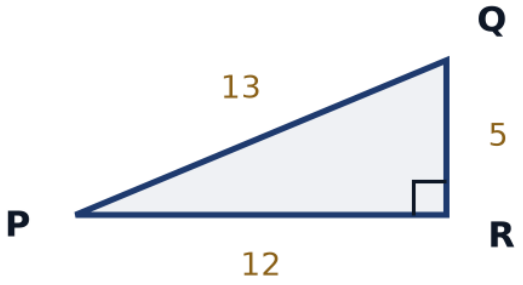
b. 19

c. 17

d. 21

24 [Foundational] • Target: Exact values in right-angled triangles

In ΔABC studying exact values in right-angled triangles, $m\angle A = 35^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 79°

b. 89°

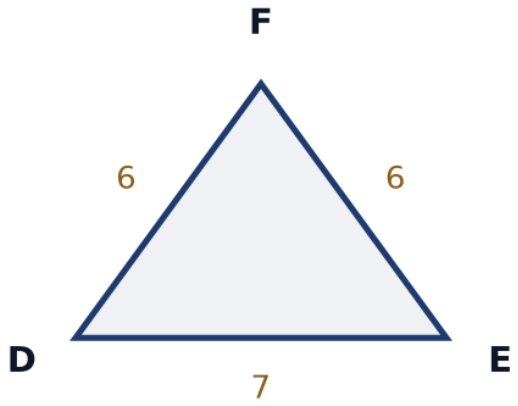
c. 71°

d. 145°

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25 [Foundational] • Target: Sine rule and the ambiguous case

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(2)$.



a. 12

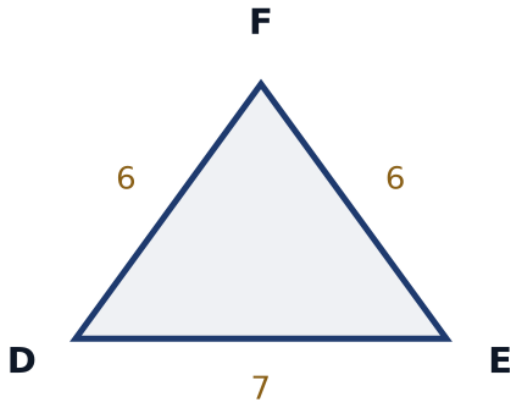
b. 15

c. 6

d. 16

26 [Foundational] • Target: Sine rule and the ambiguous case

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

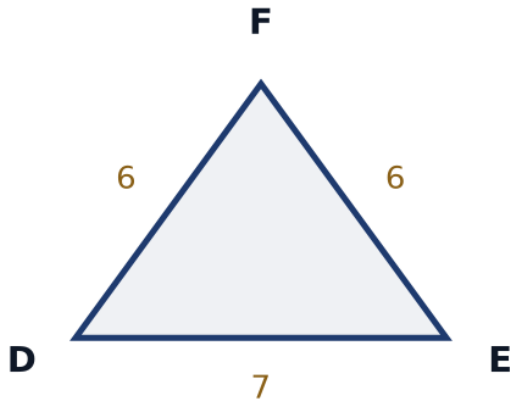
c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

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27 [Foundational] • Target: Sine rule and the ambiguous case

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(5)$.



a. 21

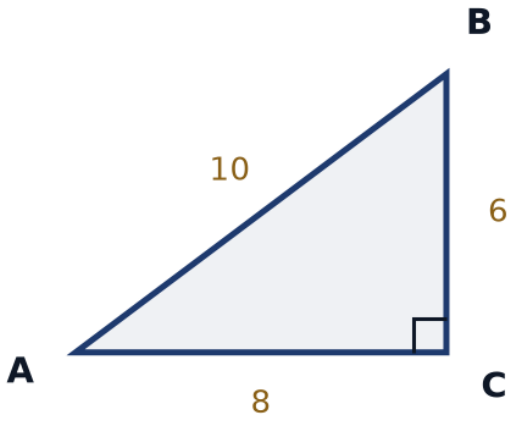
b. 24

c. 15

d. 25

28 [Foundational] • Target: The unit circle and angles of any magnitude

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 98°

b. 108°

c. 90°

d. 143°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Exact values in right-angled triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

30 [Intermediate] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 11

c. 9

d. 13

31 [Intermediate] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 9

b. 15

c. 13

d. 17

32 [Intermediate] • Target: Exact values in right-angled triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

33 [Intermediate] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 17

b. 23

c. 21

d. 25

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34 [Intermediate] • Target: Sine rule and the ambiguous case

A data set recorded for sine rule and the ambiguous case consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

35 [Intermediate] • Target: Sine rule and the ambiguous case

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(3)$.

a. 15

b. 18

c. 9

d. 19

36 [Intermediate] • Target: Sine rule and the ambiguous case

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

37 [Intermediate] • Target: The unit circle and angles of any magnitude

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 2.01 Functions

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 91°

b. 101°

c. 83°

d. 143°

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39 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 3.01 Geometry & Trigonometry

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 84°

b. 94°

c. 76°

d. 143°

40 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 4.01 Statistics & Probability

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

41 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 6.01 AI SL — Specific Topics

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 13

b. 17

c. 15

d. 17

42 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 1.01 Number & Algebra

A data set recorded for solving trigonometric equations 1 consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

43 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 2.01 Functions

For the model $f(x) = 6x + 9$ in solving trigonometric equations 1, evaluate $f(4)$.

a. 33

b. 39

c. 24

d. 37

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44 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 3.01 Geometry & Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

45 [Challenging] • Target: Solving trigonometric equations 2 • Cross-Lesson: 4.01 Statistics & Probability

For the model $f(x) = 2x + 3$ in solving trigonometric equations 2, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

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